

Name of the experiment : Verify the function of X-OR gate using basic gate:

Objective: The objective of the experiment is to verify the function of X-OR gate using basic gate.

Theory: The X-OR gate is defined as a logic gate with 2 input to perform the exclusive Disjunction operation. The main Boolean Expression of X-OR gate is,

$$Y = \bar{A}B + A\bar{B}$$

The output of 2 input X-OR gate is High only when one of the its inputs are high. If both the inputs are same then the output is low. So the X-OR circuit with 2 input is designed using AND, OR, NOT gates as shown below.

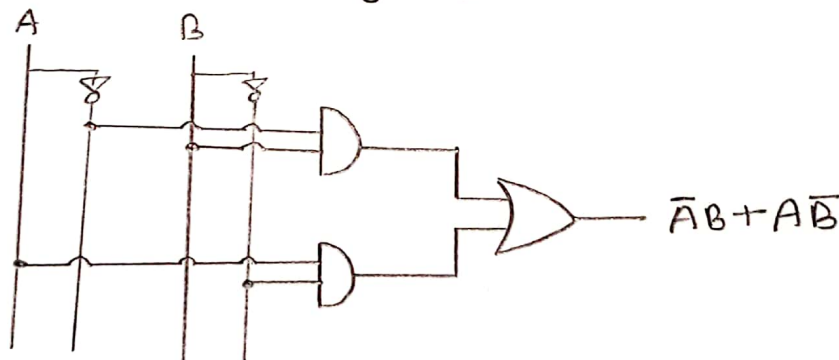


fig: circuit diagram.

Truth table

Input		output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$$Y = \bar{A}B + A\bar{B}$$

Apparatus:

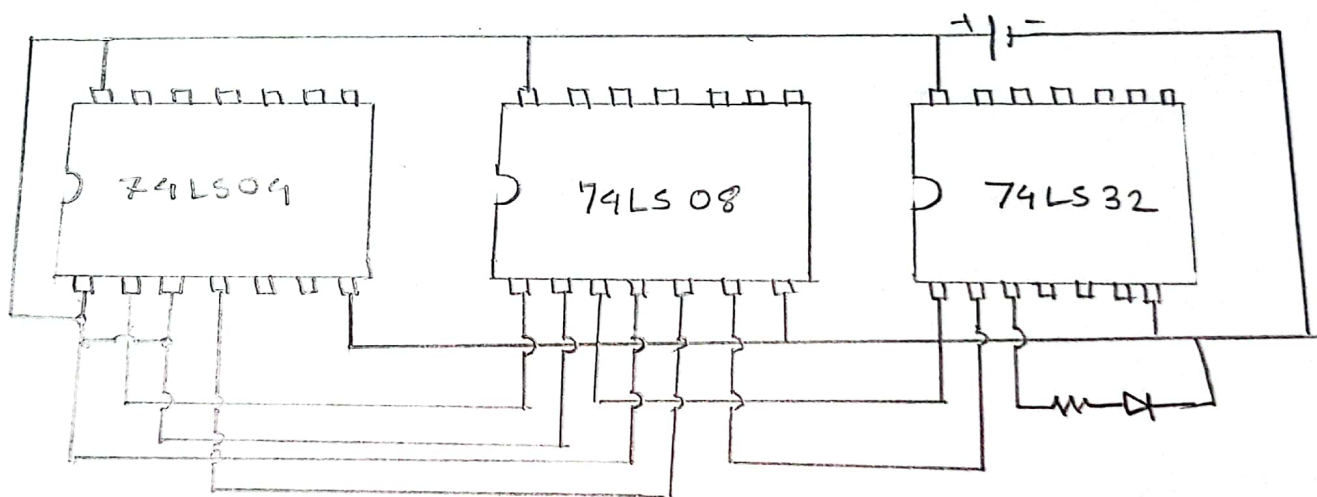
- ① Breadboard
- ② AND, OR, NOT gate
- ③ Resistor
- ④ LED
- ⑤ Power supply
- ⑥ Wire.

Procedure: At first we have to identify pin number of IC. IC are set on breadboard for X-OR gate. First, we have to make two base for Vcc and GND. There power will be connected. The first IC in the breadboard is NOT gate IC which input in pin no 1 and give output in pin 2 also pin 3 take input and 4 give output.

Here pin 1, 2 is for A and pin 3, 4 is for B. Now from gate IC pin 1 and pin 4 connect to AND gate it give $A \cdot \bar{B}$. same this way we will get output on pin 6 of AND gate which will be $\bar{A}B$. Now connect the pin 1 and 2 which will give output $\bar{A}B + A\bar{B}$ at or gates pin 3.

Working diagram:

$$\bar{A}B + A\bar{B}$$



Result

A	B	Y	LED
0	0	0	off
0	1	1	on
1	0	1	on
1	1	0	off

Discussion: We check all the gates of ICs and found the active gates. we found out the position where the IC got proper supply. For this experiment we use this IC's.